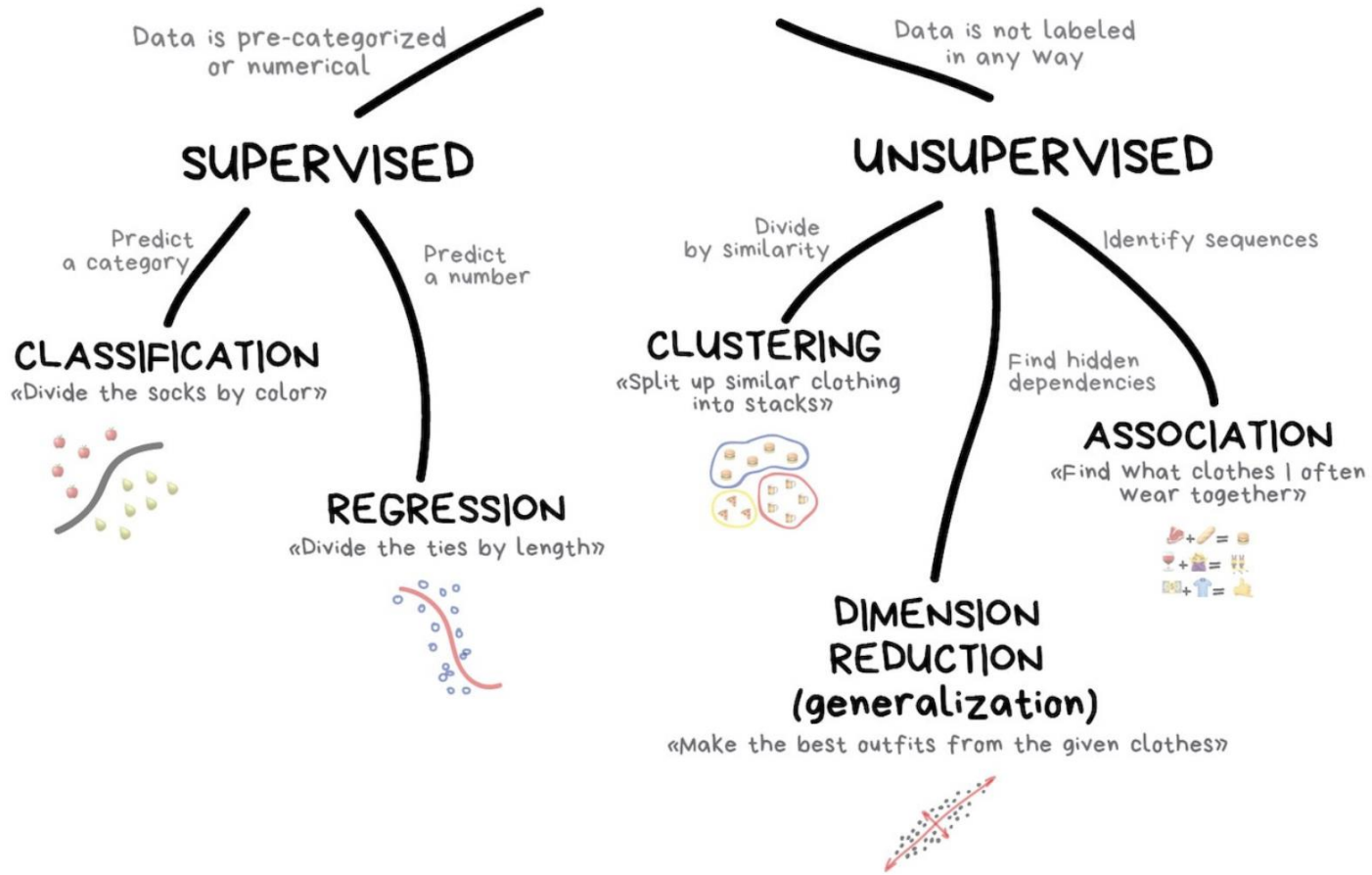
Principle Component Analysis (PCA) and Its Applications

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Classical Machine Learning



https://vas3k.com/blog/machine_learning/?ref=hn

Example – Breakfast Cereals

name	mfr	type	calories	protein	…	rating
100%_Bran	N	C	70	4	…	68
100%_Natural_Bran	Q	C	120	3	…	34
All-Bran	K	C	70	4	…	59
All-Bran_with_Extra_Fiber	K	C	50	4	…	94
Almond_Delight	R	C	110	2	…	34
Apple_Cinnamon_Cheerios	G	C	110	2	…	30
Apple_Jacks	K	C	110	2	…	33
Basic_4	G	C	130	3	…	37
Bran_Chex	R	C	90	2	…	49
Bran_Flakes	P	C	90	3	…	53
Cap'n'Crunch	Q	C	120	1	…	18
Cheerios	G	C	110	6	…	51
Cinnamon_Toast_Crunch	G	C	120	1	…	20

Description of Variables

Name: name of cereal

mfr: manufacturer

type: cold or hot

calories: calories per serving

protein: grams

fat: grams

sodium: mg.

fiber: grams

carbo: grams complex carbohydrates

sugars: grams

potass: mg.

vitamins: % FDA rec

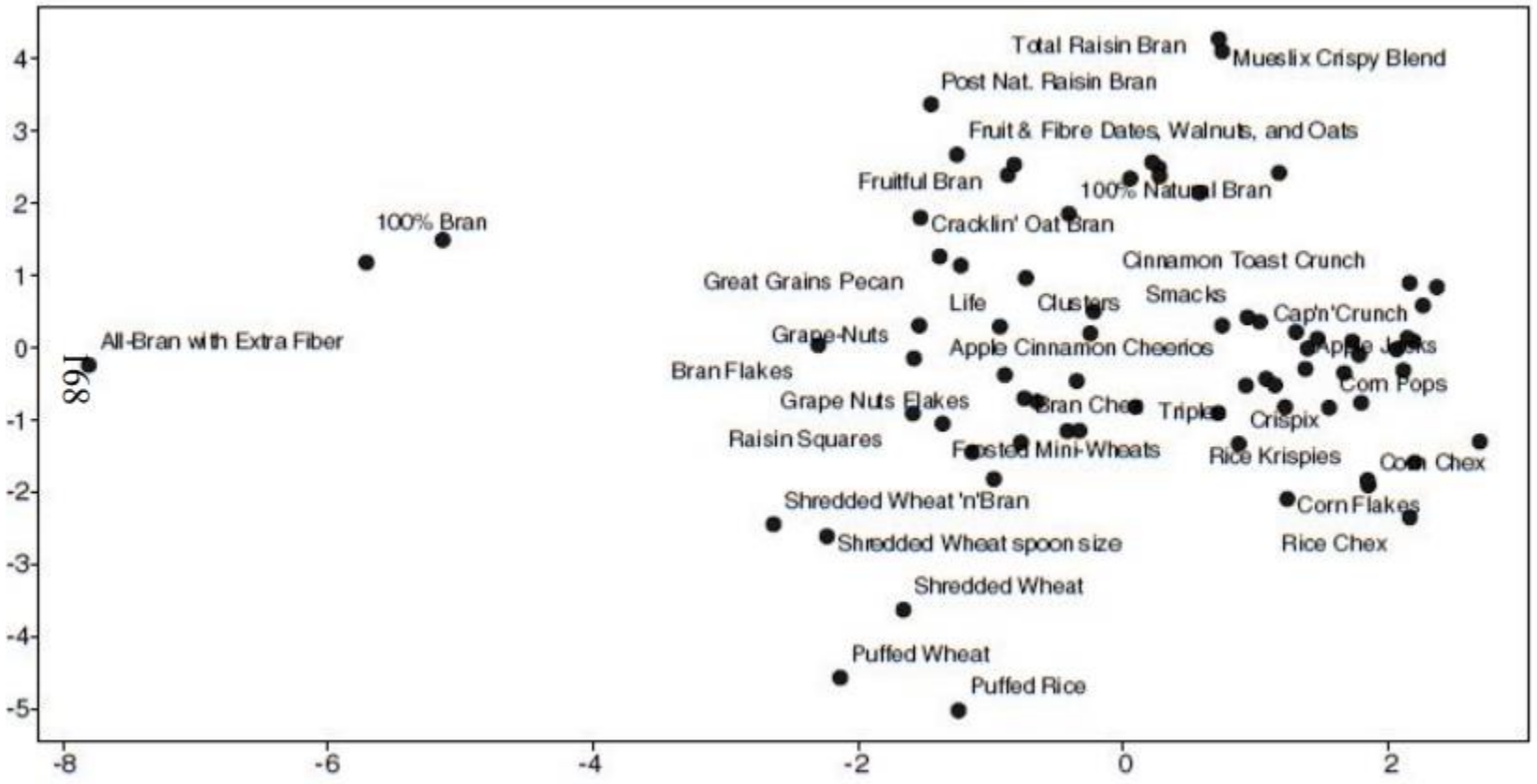
shelf: display shelf

weight: oz. 1 serving

cups: in one serving

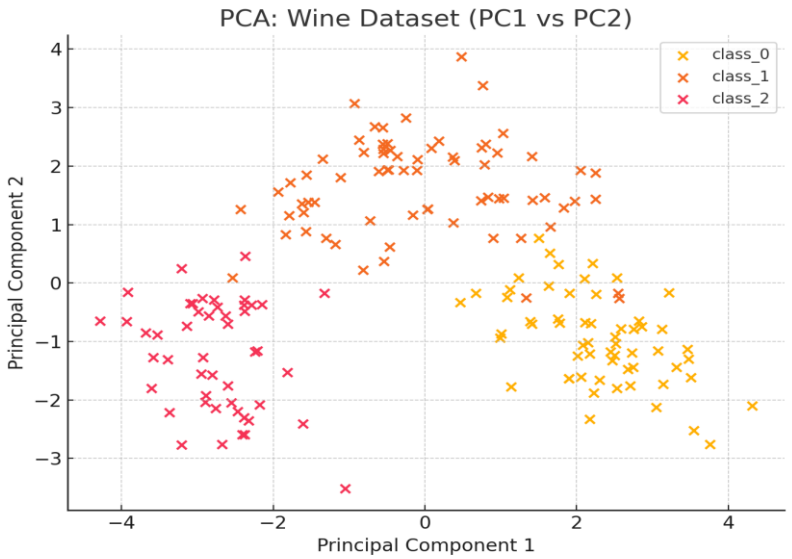
rating: consumer reports

PCA for visualization



PCA for Visualization (EDA)

- Use PCA to project data onto 2D or 3D space.
- Visualize clusters, patterns, and outliers.
- Ideal for understanding complex datasets at a glance.
- Plot PC1 vs PC2 to uncover structure in data.



Principal Components Analysis (PCA)

Goal: Reduce a set of numerical variables.

The idea: "Information" is measured by the sum of the variances of the variables.

Final product: A smaller number of numerical variables that contain most of the information

Conceptual Foundation of PCA

- Goal: Reduce high-dimensional data to fewer dimensions while retaining variance.
- PCA finds new variables (principal components) that capture maximum variance.
- These components are linear combinations of original variables.
- Principal components are uncorrelated and ranked by variance explained.

How PCA works

How does PCA do this?

- Create new variables that are linear combinations of the original variables (i.e., they are weighted averages of the original variables).

- These linear combinations are uncorrelated (no information overlap), and only a few of them contain most of the original information.

- The new variables are called *principal components*.

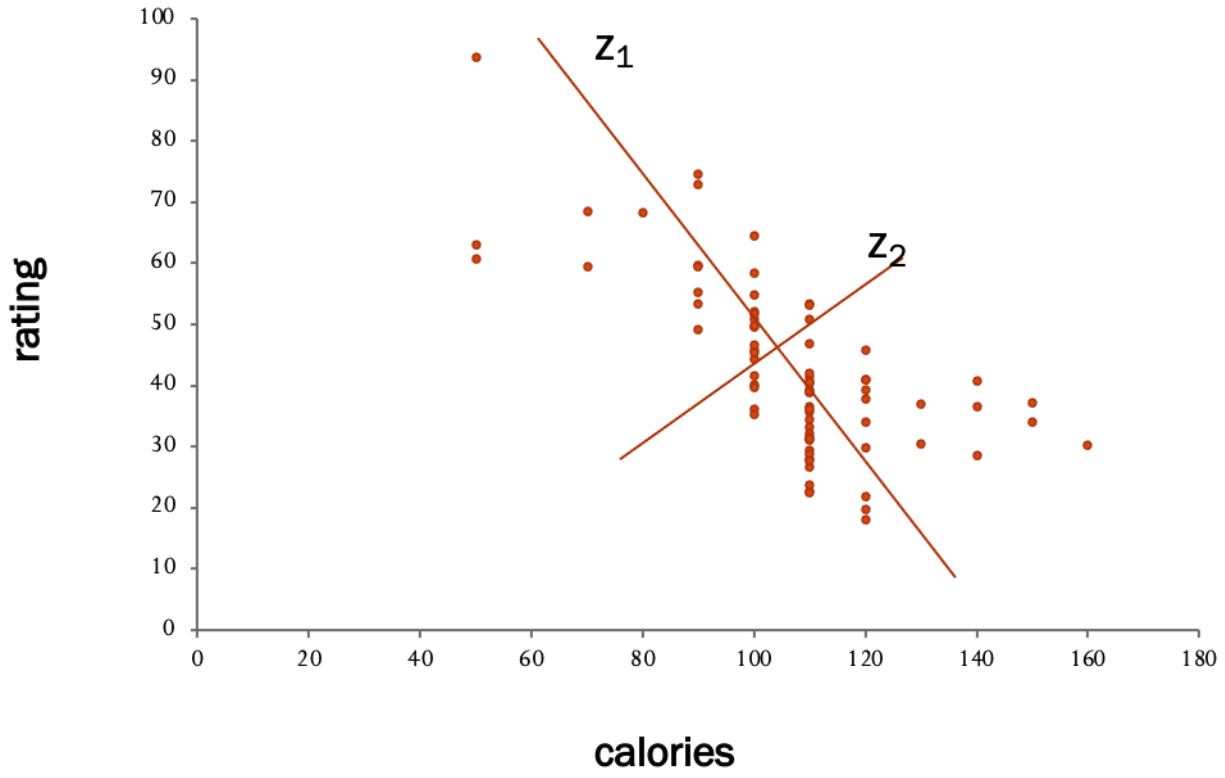
Consider calories & ratings

- Total variance (=“information”) is sum of individual variances: 379.63 + 197.32
- Calories accounts for 379.63/577 = 66%

First & Second Principal Components

Z_1 and Z_2 are two linear combinations.

- Z_1 has the highest variation (spread of values)
- Z_2 has the lowest variation



PCA output for these 2 variables

Top: weights to project original data onto z_1 & z_2

e.g. (-0.847, 0.532) are weights for z_1

Bottom: reallocated variance for new variables

z_1 : 86% of total variance

z_2 : 14%

	Components	
Variable	**1**	**2**
calories	-0.84705347	0.53150767
rating	0.53150767	0.84705347
Variance	498.0244751	78.932724
Variance%	86.31913757	13.68086338
Cum%	86.31913757	100
P-value	0	1

Principal Component Scores

Row Id.	1	2
100%_Bran	44.92	2.20
100%_Natural_Bran	-15.73	-0.38
All-Bran	40.15	-5.41
All-Bran_with_Extra_Fiber	75.31	13.00
Almond_Delight	-7.04	-5.36
Apple_Cinnamon_Cheerios	-9.63	-9.49
Apple_Jacks	-7.69	-6.38
Basic_4	-22.57	7.52
Bran_Chex	17.73	-3.51

Weights are used to compute the above scores

- e.g., col. 1 scores are computed z_1 scores using weights (-0.847, 0.532)

Properties of the resulting variables

New distribution of information:

- New variances = 498 (for z_1) and 79 (for z_2)
- Sum of variances = sum of variances for original variables *calories* and *ratings*
- New variable z_1 has most of the total variance, might be used as proxy for both *calories* and *ratings*

- z_1 and z_2 have correlation of zero (no information overlap)

Generalization

$X_1, X_2, X_3, \ldots X_p$, original *p* variables

$Z_1, Z_2, Z_3, \ldots Z_p$, weighted averages of original variables

All pairs of Z variables have 0 correlation

Order Z's by variance (z_1 largest, z_p smallest)

Usually the first few Z variables contain most of the information, and so the rest can be dropped.

PCA on full data set

Variable	1	2	3	4	5	6
calories	0.07624155	-0.01066097	0.61074823	-0.61706442	0.45754826	0.12601775
protein	-0.00146212	0.00873588	0.00050506	0.0019389	0.05533375	0.10379469
fat	-0.00013779	0.00271266	0.01596125	-0.02595884	-0.01839438	-0.12500292
sodium	0.98165619	0.12513085	-0.14073193	-0.00293341	0.01588042	0.02245871
fiber	-0.00479783	0.03077993	-0.01684542	0.02145976	0.00872434	0.271184
carbo	0.01486445	-0.01731863	0.01272501	0.02175146	0.35580006	-0.56089228
sugars	0.00398314	-0.00013545	0.09870714	-0.11555841	-0.29906386	0.62323487
potass	-0.119053	0.98861349	0.03619435	-0.042696	-0.04644227	-0.05091622
vitamins	0.10149482	0.01598651	0.7074821	0.69835609	-0.02556211	0.01341988
shelf	-0.00093911	0.00443601	0.01267395	0.00574066	-0.00823057	-0.05412053
weight	0.0005016	0.00098829	0.00369807	-0.0026621	0.00318591	0.00817035
cups	0.00047302	-0.00160279	0.00060208	0.00095916	0.00280366	-0.01087413
rating	-0.07615706	0.07254035	-0.30776858	0.33866307	0.75365263	0.41805118

Variance	7204.161133	4833.050293	498.4260864	357.2174377	72.47863007	4.33980322
Variance%	55.52834702	37.25226212	3.84177661	2.75336623	0.55865192	0.0334504
Cum%	55.52834702	92.78060913	96.62238312	99.37575531	99.93440247	99.96785736

- First 6 components shown
- First 2 capture 93% of the total variation
- Note: data differ slightly from text

Normalizing data

- In these results, sodium dominates first PC
- Just because of the way it is measured (mg), its scale is greater than almost all other variables
- Hence its variance will be a dominant component of the total variance
- Normalize each variable to remove scale effect

  Divide by std. deviation (may subtract mean first)
- Normalization (= standardization) is usually performed in PCA; otherwise measurement units affect results
- Note: In RapidMiner, use correlation matrix option to use normalized variables

PCA using standardized variables

Variable	1	2	3	4	5	6
calories	0.32422706	0.36006299	0.13210163	0.30780381	0.08924425	-0.20683768
protein	-0.30220962	0.16462311	0.2609871	0.43252215	0.14542894	0.15786675
fat	0.05846959	0.34051308	-0.21144024	0.37964511	0.44644874	0.40349057
sodium	0.20198308	0.12548573	0.37701431	-0.16090299	-0.33231756	0.6789462
fiber	-0.43971062	0.21760374	0.07857864	-0.10126047	-0.24595702	0.06016004
carbo	0.17192839	-0.18648526	0.56368077	0.20293142	0.12910619	-0.25979191
sugars	0.25019819	0.3434512	-0.34577203	-0.10401795	-0.27725372	-0.20437138
potass	-0.3834067	0.32790738	0.08459517	0.00463834	-0.16622125	0.022951
vitamins	0.13955688	0.16689315	0.38407779	-0.52358848	0.21541923	0.03514972
shelf	-0.13469705	0.27544045	0.01791886	-0.4340663	0.59693497	-0.12134896
weight	0.07780685	0.43545634	0.27536476	0.10600897	-0.26767638	-0.38367996
cups	0.27874646	-0.24295618	0.14065795	0.08945525	0.06306333	0.06609894
rating	-0.45326898	-0.22710647	0.18307236	0.06392702	0.03328028	-0.16606605

Variance	3.59530377	3.16411042	1.86585701	1.09171081	0.96962351	0.72342771
Variance%	27.65618324	24.3393116	14.35274601	8.39777565	7.45864248	5.5648284
Cum%	27.65618324	51.99549484	66.34824371	74.74601746	82.20465851	87.76948547

- First component accounts for smaller part of variance
- Need to use more components to capture same amount of information

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- High scores on principal **component-1** mean that the cereal 166 is high in calories and the amount per bowl, and low in protein, fiber, and potassium. And, with a low consumer rating.
- The **component-2** is most affected by the weight of a serving,
- The **component-3** most affected by the carbohydrate content. And ….

Wine Data

ExampleSet (178 examples, 0 special attributes, 14 regular attributes)

Filter (178 / 178 examples): all

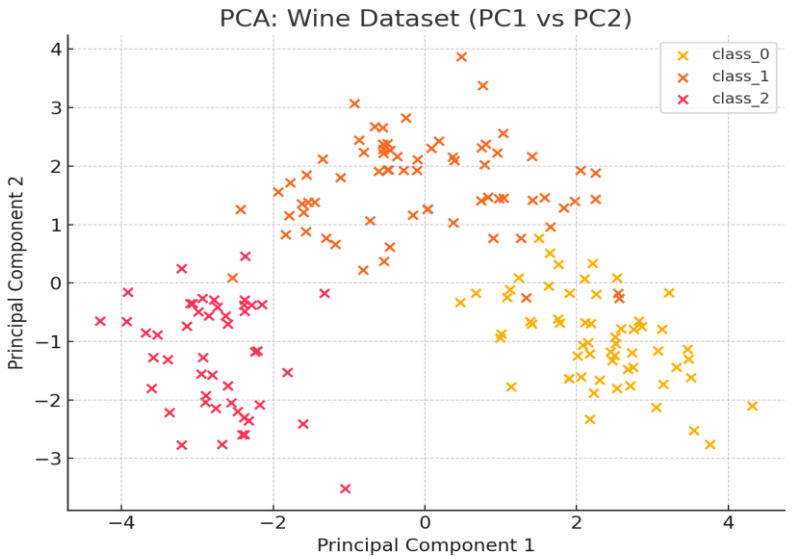
..	Alcohol	Malic_Acid	Ash	Ash_...	Magnesium	Total_Phen...	Flavanoi...	Nonflavanoi...	Proanthocy...	Color_Int...	Hue	OD2...	Proline	Custom
.	14.230	1.710	2.430	15.600	127	2.800	3.060	0.280	2.290	5.640	1.040	3.920	1065	1
.	13.200	1.780	2.140	11.200	100	2.650	2.760	0.260	1.280	4.380	1.050	3.400	1050	1
.	13.160	2.360	2.670	18.600	101	2.800	3.240	0.300	2.810	5.680	1.030	3.170	1185	1
.	14.370	1.950	2.500	16.800	113	3.850	3.490	0.240	2.180	7.800	0.860	3.450	1480	1
.	13.240	2.590	2.870	21	118	2.800	2.690	0.390	1.820	4.320	1.040	2.930	735	1
.	14.200	1.760	2.450	15.200	112	3.270	3.390	0.340	1.970	6.750	1.050	2.850	1450	1
.	14.390	1.870	2.450	14.600	96	2.500	2.520	0.300	1.980	5.250	1.020	3.580	1290	1
.	14.060	2.150	2.610	17.600	121	2.600	2.510	0.310	1.250	5.050	1.060	3.580	1295	1
.	14.830	1.640	2.170	14	97	2.800	2.980	0.290	1.980	5.200	1.080	2.850	1045	1
.	13.860	1.350	2.270	16	98	2.980	3.150	0.220	1.850	7.220	1.010	3.550	1045	1
.	14.100	2.160	2.300	18	105	2.950	3.320	0.220	2.380	5.750	1.250	3.170	1510	1
.	14.120	1.480	2.320	16.800	95	2.200	2.430	0.260	1.570	5	1.170	2.820	1280	1
.	13.750	1.730	2.410	16	89	2.600	2.760	0.290	1.810	5.600	1.150	2.900	1320	1
.	14.750	1.730	2.390	11.400	91	3.100	3.690	0.430	2.810	5.400	1.250	2.730	1150	1
.	14.380	1.870	2.380	12	102	3.300	3.640	0.290	2.960	7.500	1.200	3	1547	1
.	13.630	1.810	2.700	17.200	112	2.850	2.910	0.300	1.460	7.300	1.280	2.880	1310	1

Wine Data with new dimensions

	Principal Components					Std
	1	2	3	4	5	Dev
Alcohol	0.001	0.013	0.014	-0.030	0.129	0.8
MalicAcid	-0.001	0.009	0.167	-0.427	-0.402	1.2
Ash	0.000	-0.002	0.054	-0.009	0.006	0.3
Ash_Alcalinity	-0.004	-0.045	0.976	0.176	0.060	3.6
Magnesium	0.014	-0.998	-0.040	-0.031	0.006	14.7
Total Phenols	0.001	0.002	-0.015	0.164	0.316	0.7
Flavanoids	0.002	0.000	-0.049	0.214	0.545	1.1
Nonflavanoid_Phenols	0.000	0.002	0.004	-0.025	-0.040	0.1
Proanthocyanins	0.001	-0.007	-0.031	0.082	0.244	0.7
Color Intensity	0.002	0.022	0.097	-0.804	0.536	1.6
Hue	0.000	-0.002	-0.021	0.096	0.064	0.2
OD280/OD315	0.001	-0.002	-0.022	0.220	0.261	0.7
Proline	1.000	0.014	0.004	0.001	-0.004	351.5
Variance	123594.453	194.345	11.424	2.388	1.391	
% Variance	99.830%	0.157%	0.009%	0.002%	0.001%	
Cumulative %	99.830%	99.987%	99.996%	99.998%	99.999%	

PCA for Visualization (EDA)

- Use PCA to project data onto 2D or 3D space.
- Visualize clusters, patterns, and outliers.
- Ideal for understanding complex datasets at a glance.
- Plot PC1 vs PC2 to uncover structure in data.



PCA for Clustering

- PCA reduces dimensionality before clustering (e.g., k-means).
- Removes noise and correlated features.
- Improves cluster separation and speed.
- Compare clustering with and without PCA.

Alternative 1: PCA Before K-Means

- 1. Standardize the data.
- 2. Apply PCA to reduce dimensions (e.g., to 2 or 3 components).
- 3. Perform K-Means clustering on the reduced dataset.
- 4. Visualize and analyze clusters in PCA space.
- ✅ Helps reduce noise and improves clustering performance.
- ✅ Easier to visualize clusters.
- ⚠️ May lose some original feature interpretability.

Alternative 2: PCA After K-Means

- 1. Standardize the data.
- 2. Perform K-Means clustering on the original high-dimensional data.
- 3. Use PCA to reduce dimensions of the data **after clustering**.
- 4. Project clusters into PCA space for visualization.
- ✅ Clustering uses full feature space (may preserve structure).
- ✅ PCA helps explain and visualize each cluster's meaning.
- ⚠️ Less useful for noise reduction before clustering.

PCA for Classification

- Apply PCA to training data to reduce feature space.
- Use selected PCs as input to classifiers.
- Reduces overfitting and simplifies decision boundaries.
- Be cautious: some predictive power might be lost.

Best Practices & Pitfalls

- Standardize data before applying PCA.
- Interpretability of PCs can be difficult.
- Don't always rely on first two PCs—check explained variance.
- Combine PCA with domain knowledge for best results.